1. (a) What are the limitations of non-conventional energy sources?

Intermittency and Variability: Higher Initial Costs: Land and Space Requirements: Visual and Noise Impacts: Environmental Footprint: Limited Resource Availability: Geographic Constraints Technological Challenges:

(b) What do you understand by greenhouse effect and from where it is generated?

The greenhouse effect is a natural process that warms the Earth's surface. It's caused by greenhouse gases in the atmosphere, such as water vapor, carbon dioxide, methane, and nitrous oxide, which trap outgoing infrared radiation (heat) after it has been radiated from the Earth's surface. This trapped heat keeps the planet's average temperature at a livable level. The greenhouse effect is generated primarily by the absorption and re-emission of infrared radiation by the atmosphere.

(c) Comment on the prospects of fossil fuels in India.

The prospects of fossil fuels in India are declining due to a growing global shift towards renewable energy, coupled with India's own commitment to reducing emissions and transitioning to a low-carbon economy. While fossil fuels continue to be a major energy source, their dominance is expected to diminish as renewable energy technologies become more cost-competitive and India invests heavily in renewable energy infrastructure.

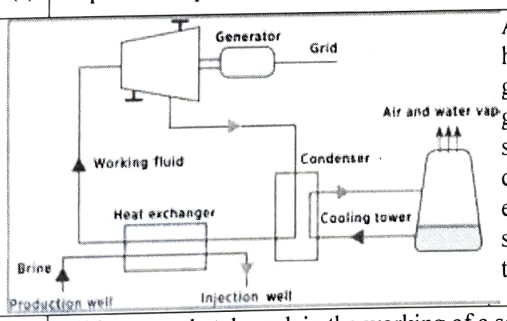
2. (a) With a neat diagram explain the operation of a heat pump.

A heat pump uses a refrigerant to transfer heat from a low-temperature source to a higher-temperature destination, effectively heating or cooling a space. The cycle involves compression, cooling, expansion, and heating, with the refrigerant acting as a heat carrier.

(b) What do you understand by the heating value of fuel?

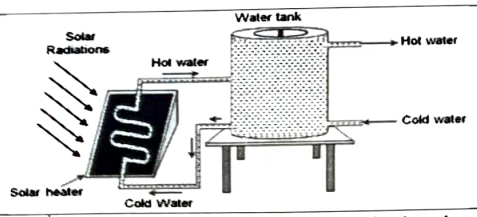
The heating value, or calorific value, of a fuel is the amount of heat energy released when a specified quantity of the fuel is completely burned in the presence of oxygen. It essentially measures how much energy is stored within the fuel and is a crucial factor in determining fuel efficiency and suitability for various applications.

(c) Explain the operation of binary cycle power plant.



A binary cycle power plant uses a geothermal heat source and a secondary working fluid to generate electricity. The heat from the geothermal source is used to vaporize the secondary fluid, and this vapor drives a turbine connected to a generator, producing electricity. The geothermal fluid and the secondary fluid are kept separate throughout the process, preventing contamination.

3. (a) With a neat sketch explain the working of a solar water heater.



A solar water heater utilizes solar energy to heat water by circulating it through black, heat-absorbing panels (solar collectors) on the roof. The heated water is then stored in an insulated tank for later use.

(b) Define declination angle and calculate its value on May 25?

N=145
declination angle=23.018

(c) Differentiate between cylindrical concentrating and solar Fresnel lens collectors.

Cylindrical concentrating lens collectors and Fresnel lens collectors are both types of concentrating solar collectors, but they differ in their design and how they focus sunlight. Cylindrical lenses focus sunlight into a line, while Fresnel lenses focus sunlight onto a focal point or area. This difference impacts their application and efficiency.

Cylindrical Concentrating Lens Collectors:	Solar Fresnel Lens Collectors:
Focusing: These lenses focus sunlight into a line, ideal for applications like laser scanning, vision quality control, and lighting. **Efficiency:** They can achieve high concentration ratios, with some designs reaching up to ten. **Applications:** They are used in laser scanning, vision quality control, metrology, ophthalmology, defense, and lighting.	**Focusing:** Fresnel lenses are known for their ability to focus sunlight onto a focal point or area. **Efficiency:** They offer high optical efficiency, making them popular for solar concentrators. **Applications:** They are widely used as solar concentrators, gathering sunlight for solar panels and other applications.

4. (a) Calculate the declination angle of beam radiation on a photovoltaic array located in Thane (25^0 10' N, 78^0 61' E) Given the following data:

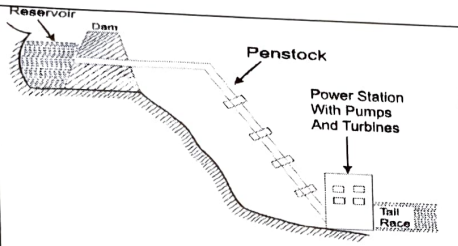
Date :October 25,2007
Time :10.30h(LAT)
Slope of PV array :18^0
Surface azimuthal angle : 27^0

N=298, delta=-13.122,beta=18,γ =27,ω=56.4,φ=25.16,θ=52.914

(b) Calculate the latitude angle for Question 4(a). **φ=25.16**

(c) Calculate the incidence angle for Question 4(a). **θ=52.914**

5. (a) Explain the low head hydro plant with diagram.



A low-head hydro plant harnesses water flow for electricity generation where the water head (vertical distance between intake and discharge points) is relatively low, typically less than 20 meters. These plants often use large, slower-speed turbines, like Kaplan or Francis turbines, to efficiently extract energy from the high volume of water flowing at a lower head.

(b) Write the key equation for the power from a hydro power plant. Write the factors on which this power depends.

The power from a hydropower plant is calculated by $P = \rho * g * Q * H * \eta$,

where: **P**: is the power output in Watts (W).
ρ: is the density of water (approximately 1000 kg/m³).
g: is the acceleration due to gravity (approximately 9.81 m/s²).
Q: is the volumetric flow rate of water (m³/s).
H: is the head or height of the water fall (m).
η: is the overall efficiency of the plant (dimensionless).

(c) Explain the future prospective of the renewable energy.

The future of renewable energy looks promising, with projections indicating significant growth in the coming years. Renewable energy sources are expected to surpass fossil fuels in electricity generation, contributing significantly to decarbonizing the power sector and mitigating climate change. Key Future Trends:

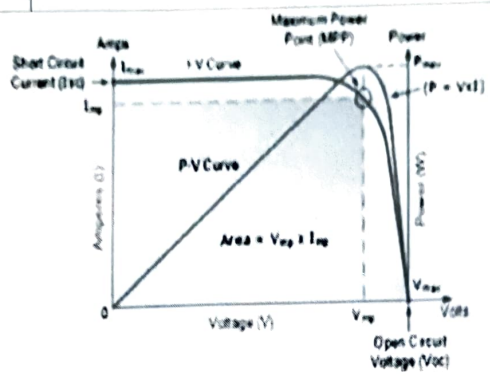
Growth in Renewable Capacity:
Technological Advancements:
Shift from Fossil Fuels:
Economic Benefits:
Policy Support:

6. (a) Explain the I-V characteristics of a solar cell and define fill factor.

The I-V characteristics of a solar cell describe the relationship between the current and voltage produced when the cell is illuminated. This relationship is crucial for understanding

the cell's performance and output power.

The fill factor (FF) is a measure of the quality of a solar cell, indicating how closely the cell's maximum power output approaches the theoretical maximum power based on its open-circuit voltage (Voc) and short-circuit current (Isc).



I-V Curve: The I-V curve plots the current produced by a solar cell against the voltage across it under different illumination conditions.

Open-Circuit Voltage (Voc): Short-Circuit Current (Isc):

Maximum Power Point (MPP): Fill Factor (FF):

FF = Pmax / (Voc * Isc)

A higher fill factor indicates a better-performing solar cell, as it produces a larger fraction of the theoretical maximum power. Typical fill factors for silicon solar cells are around 70-80%.

(b) What is the importance of MPPT in an SPV system?

In a standalone photovoltaic (SPV) system, MPPT (Maximum Power Point Tracking) is crucial for maximizing the energy harvested from solar panels. It ensures the solar panels operate at their peak output by dynamically adjusting the voltage and current to match changing conditions like sunlight intensity and temperature. This optimization results in a more efficient and reliable energy supply, especially in fluctuating conditions like cloudy days or partial shading.

Here's a more detailed breakdown of MPPT's importance:

Optimized Power Output:

Increased Energy Harvest:

Improved Efficiency:

Adaptability to Changing Conditions:

Battery Charging Optimization:

System Performance:

(c) What do you understand by indirect solar energy?

Indirect solar energy refers to forms of energy that are derived from solar radiation but are not directly harnessed from sunlight itself. Instead, they are converted from solar energy into other energy forms like chemical, wind, or tidal energy. Examples of indirect solar energy include biomass, wind power, and hydroelectricity.

7. (a) The cost of major components of solar thermal power plant is specified in the following table along with the life spans.

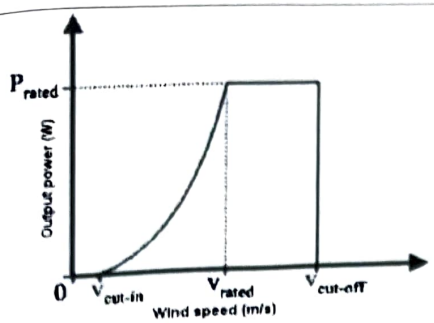
Item	Cost in s/KW	Life Period
Heat energy collectors	22000	20
Boiler + steam turbine	11900	10
Electric Generator	6500	10
Accessories and tools	500	5

If the rate of interest is 15% and the maintenance cost is 2% of the capital cost per year. Calculate the capital cost per KW. 40900

(b) Calculate the maintenance cost per KW. 8929.47

(c) Calculate life cycle cost (LCC) per KW. 49829.47

		7 Capital cost = 22000 + 11900 + 6500 + 500 = 40900 Maintaince Cost = $\frac{2}{100} \times 40900 \left[\frac{1}{0.15} \left(1 - \frac{1}{1+0.15} \right)^1 \right]$ $= 818 \left[83.33 \left(1 - \frac{1}{0.869} \right) \right]$ $= 818 \times 10.916 = 8929.47$ Replacement cost = 0 LCC = 40900 + 8929.47 = 49829.44
8.	(a)	Discuss various types of drive schemes used in wind turbines.
		Wind turbine drive schemes primarily differ in how the rotating energy from the rotor is converted into electricity and whether a gearbox is used. The main types are fixed-speed, direct-drive, and variable-speed (using a doubly-fed induction generator or a permanent magnet generator), with each having its own advantages and disadvantages.
		1. Fixed-Speed Systems: **Description:** These systems use a gearbox to increase the rotor's low speed to match the generator's speed, which is then directly connected to the grid. **Advantages:** Simpler design, lower cost, and more mature technology. **Disadvantages:** Less efficient in capturing wind energy at varying wind speeds, as the turbine must operate at a constant speed. 2. Direct-Drive Systems: **Description:** The generator is directly connected to the rotor, eliminating the gearbox. **Advantages:** Higher reliability and fewer mechanical components, potentially lower maintenance. **Disadvantages:** Higher cost, requires larger and slower generators.
		3. Variable-Speed Systems: **Description:** These systems allow the turbine to operate at varying speeds to maximize energy capture. **Advantages:** Higher energy capture, especially in variable wind conditions. **Disadvantages:** More complex design, requires power electronics to manage the variable speed. 3.1 Doubly-Fed Induction Generators (DFIG): 3.2 Permanent Magnet Generators (PMSG): 4. Hybrid Systems: **Description:** Combine aspects of both fixed-speed and variable-speed systems, such as using a gearbox with a variable-speed generator. **Advantages:** Can provide a balance between efficiency and cost. **Disadvantages:** Can be more complex than either fixed or variable-speed systems.
	(b)	Explain the yaw control method in wind energy.
		Yaw control in wind energy refers to the system that aligns a wind turbine's nacelle and rotor with the wind direction, maximizing the amount of wind energy captured. This is crucial for horizontal axis wind turbines (HAWTs) as it ensures the blades are facing the wind, maximizing their efficiency. 1. Purpose of Yaw Control: • **Maximizing Power Output:** • **Wind Direction Changes:**
	(c)	What do you understand by offshore wind firm, explain.
		An offshore wind farm is a collection of wind turbines located in bodies of water, typically the ocean or large lakes, to harness wind energy and generate electricity. They are essentially power plants that utilize the wind's kinetic energy to produce electricity, which is then transmitted back to shore. Offshore wind farms offer several advantages, including: **Higher wind speeds:** **Less land use:** **Reduced visual impact:**
9.	(a)	With the help of a diagram, discuss the power versus wind speed characteristics of a wind turbine.



A wind turbine's power output increases with wind speed, but not linearly. It's a characteristic curve with a cut-in speed, a point where power generation begins, and a rated speed, where maximum power output is achieved. Beyond the rated speed, the turbine may stall or furl to protect against damage.

Explanation:

1. Cut-in Speed:

2. Increasing Power Output:

3. Rated Speed:

(b) Discuss stall control used in wind turbines.

Stall control in wind turbines limits power output by allowing the blades to stall, reducing lift and torque at high wind speeds. This is achieved by passively designed blades or, in some cases, actively forcing a stall using blade pitch adjustments.

Passive Stall Control: Blades are designed with a fixed pitch, and their shape is aerodynamically engineered to stall at high wind speeds. This method is simple and cost-effective but has limited control and can lead to torque fluctuations.

Active Stall Control: Blades are actively rotated to induce a stall, similar to how pitch control adjusts blade angles. This allows for more precise control over power output but requires a more complex system.

Advantages: Simple, low cost (passive), and effective at limiting overspeed during gust conditions. **Disadvantages:**Limited control, potential for torque fluctuations (passive), and more complex (active).

(c) What do you understand by energy farming?

Energy farming encompasses practices where crops are specifically cultivated to produce energy, such as biofuels or for direct combustion. It also involves utilizing energy-efficient techniques in farming operations, like renewable energy sources and optimized irrigation. Essentially, energy farming aims to both produce energy from agricultural resources and reduce the energy intensity of farming practices.

Here's a more detailed look:

1. Energy Crops:

2. Energy-Efficient Farming Practices:

Efficient irrigation:

Renewable energy:

Optimized machinery:

Biogas production:

3. Benefits of Energy Farming:

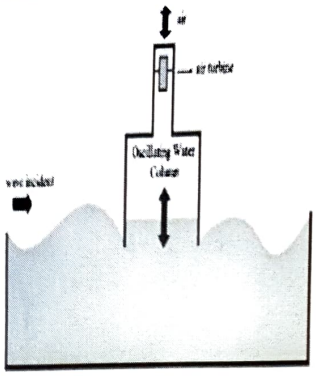
Renewable energy source:

Environmental benefits:

Economic opportunities:

Reduced energy consumption:

10 (a) Explain the operation of oscillating water type of wave device.



Oscillating Water Column (OWC) wave energy devices capture energy from ocean waves by using the rise and fall of a water column to compress and decompress air within a chamber, which then drives a turbine and generates electricity. This is achieved by a partially submerged hollow structure open to the sea below the waterline, containing a column of trapped air above the water column.

Wave Interaction:

Air Compression and Expansion:

Air Turbine Activation:

Electricity Generation:

OWC devices can be either fixed or floating, and they can be located onshore or offshore in deeper waters. The turbine may also be connected to a generator to produce electricity.

(b) Explain various types of geothermal resources.

Geothermal resources are naturally heated water, steam, or hot rocks beneath the Earth's surface. They are classified into several types based on their temperature, pressure, and how they are used for energy production. The primary types include hydrothermal reservoirs, geo pressurized resources, hot dry rock, and magma. These resources can be used for direct heating, electricity generation, and geothermal heat pumps.

	(c)	What do you understand by tidal barrage and tidal basin?
		A tidal barrage is a dam-like structure built across the mouth of a bay or estuary to capture the energy of the tides. A tidal basin is a body of water created by the barrage, often enclosed by a dam, where water can be trapped and released to generate electricity.

	Tidal Barrage: It's a dam-like structure built across the mouth of a bay or estuary. It traps water during high tide, creating a difference in water levels on either side of the barrage. This difference in water level is used to generate electricity as water flows through turbines.	Tidal Basin: It's the area enclosed by the barrage. Water is allowed to flow into the basin on the incoming tide and out on the outgoing tide, creating a controlled flow of water. The controlled flow of water through the turbines generates electricity.